DEEP LEARNING

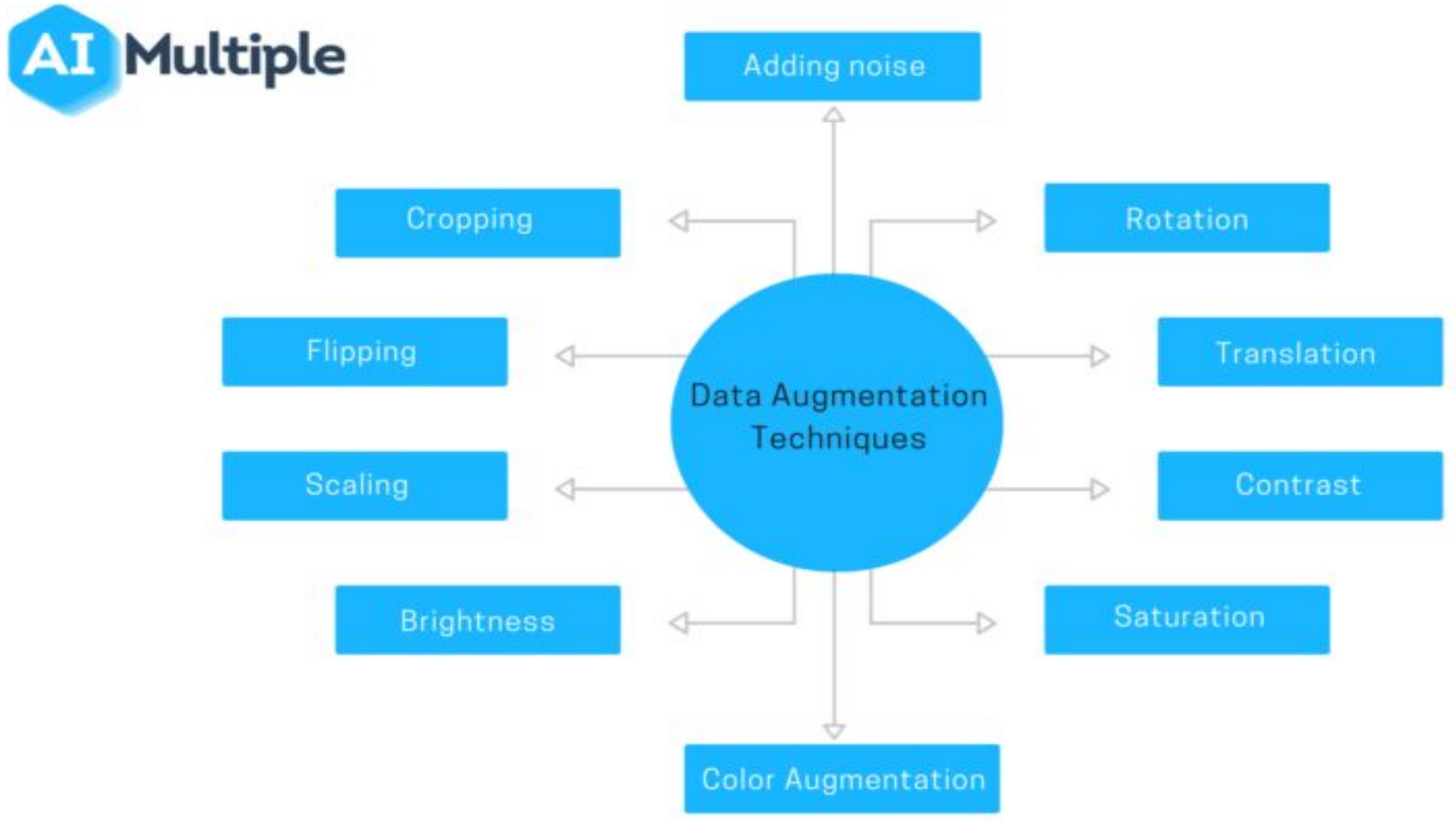
Unit – 5

Computer Science and Engineering
RV College of Engineering
Bengaluru – 59

Data Augmentation

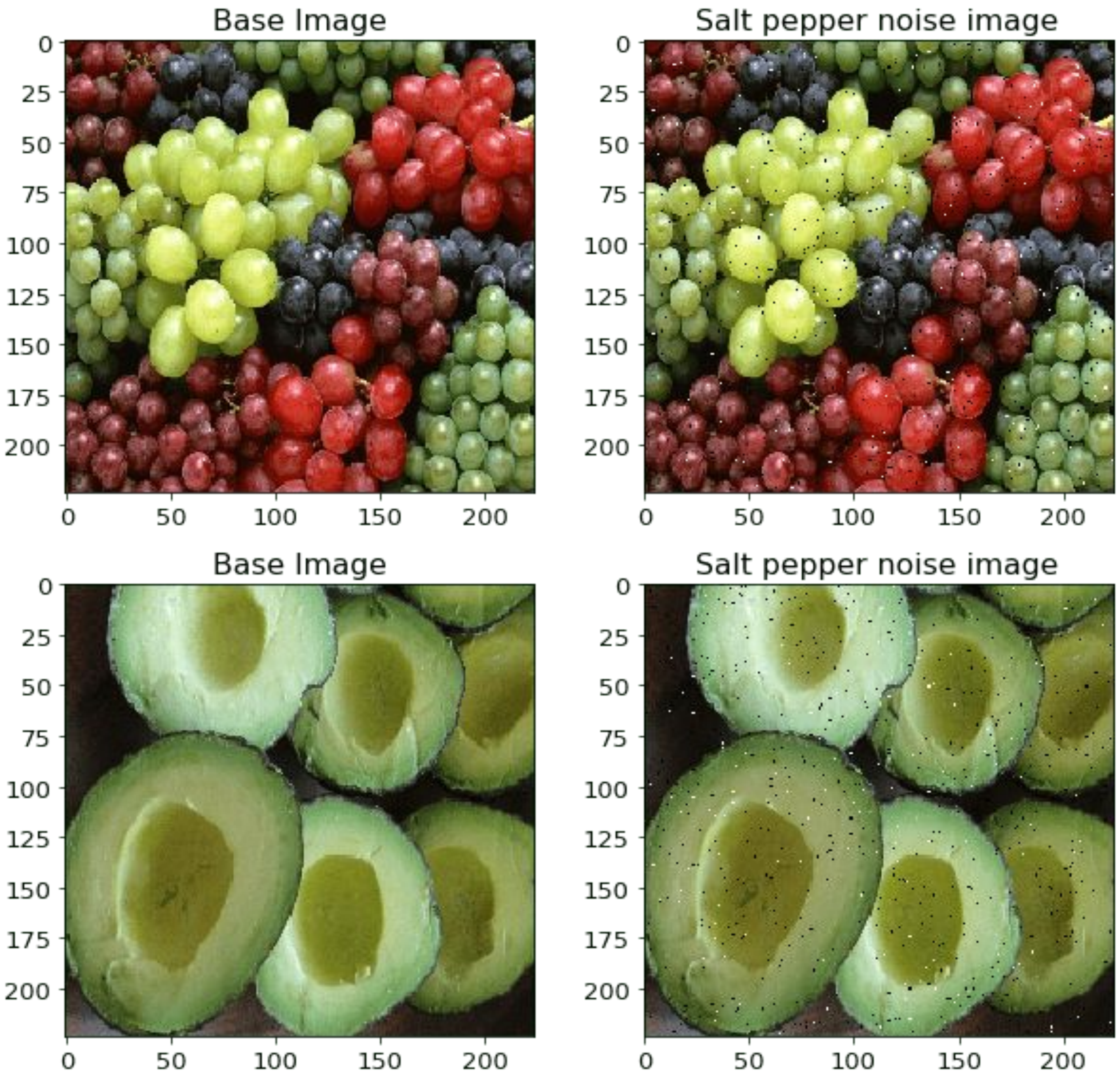
- Data augmentation techniques generate different versions of a real dataset artificially to increase its size
- Computer vision and natural language processing (NLP) models use data augmentation strategy to handle with data scarcity and insufficient data diversity
- Data augmentation algorithms can increase accuracy of machine learning models

Data Augmentation



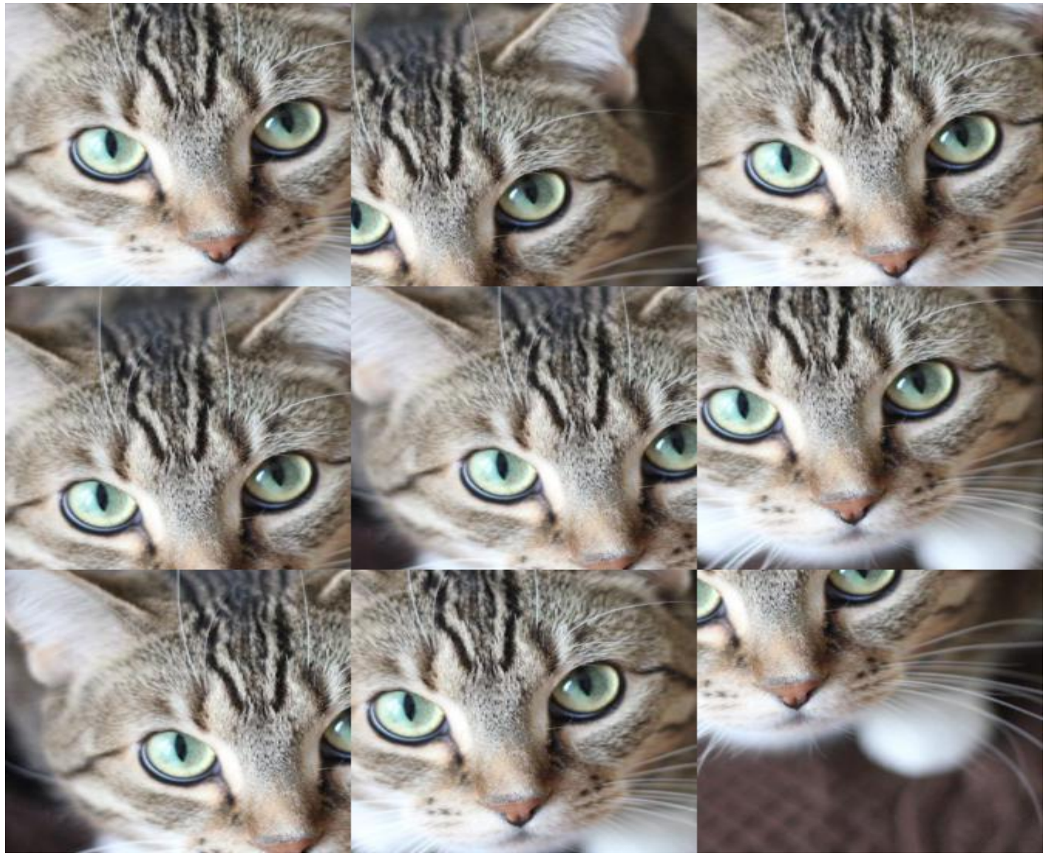
Data Augmentation

- Adding noise
- For blurry images, adding noise on the image can be useful.
- By “salt and pepper noise”, the image looks like consisting of white and black dots.



Data Augmentation

- Cropping
- A section of the image is selected, cropped and then resized to the original image size.



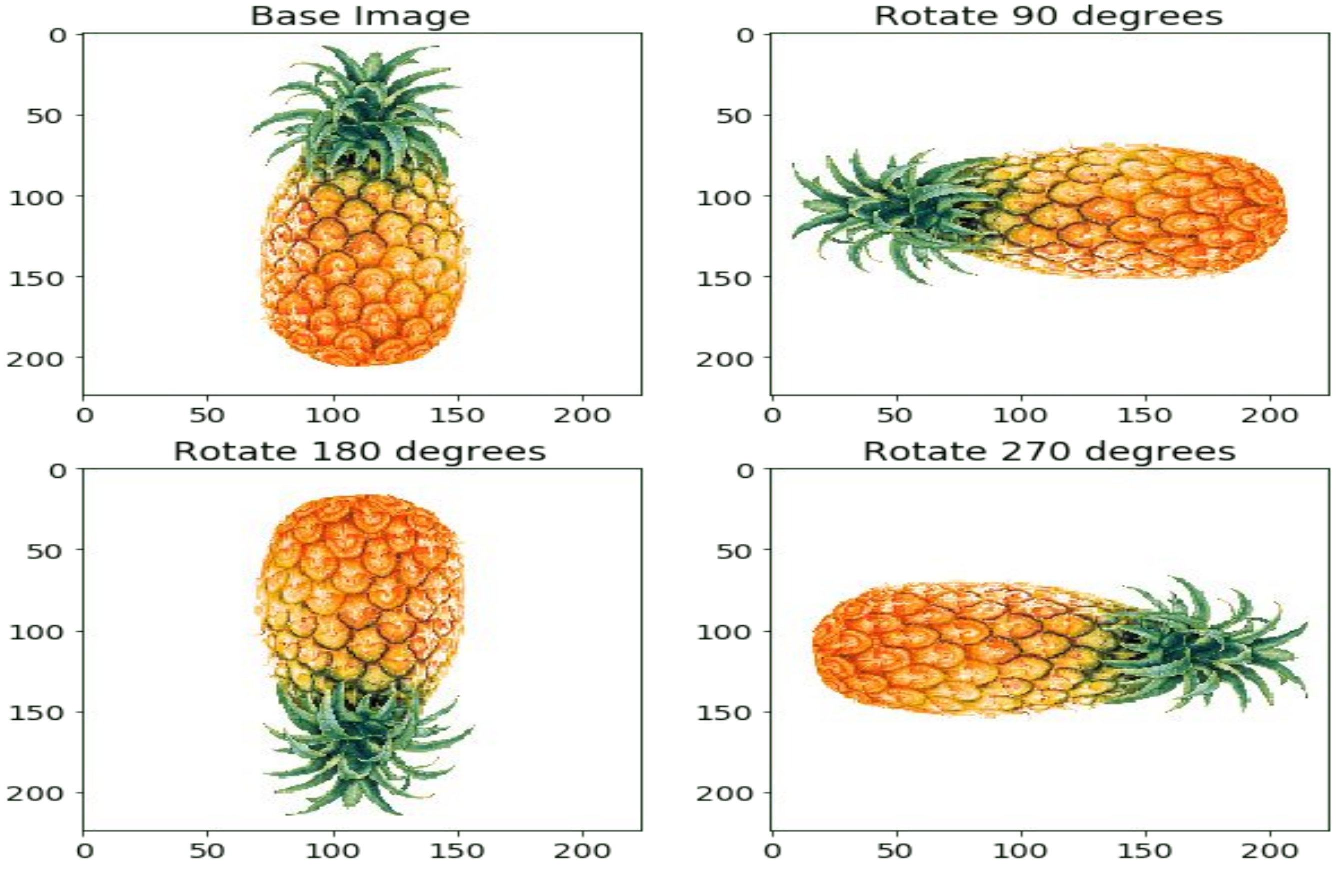
Data Augmentation

- Flipping
- The image is flipped horizontally and vertically.
- Flipping rearranges the pixels while protecting the features of the image.
- Vertical flipping is not meaningful for some photos, but it can be useful in cosmology or for microscopic photos.

1:ORIGINAL IMAGE – 2. HORIZONTAL FLIP – 3.VERTICAL FLIP

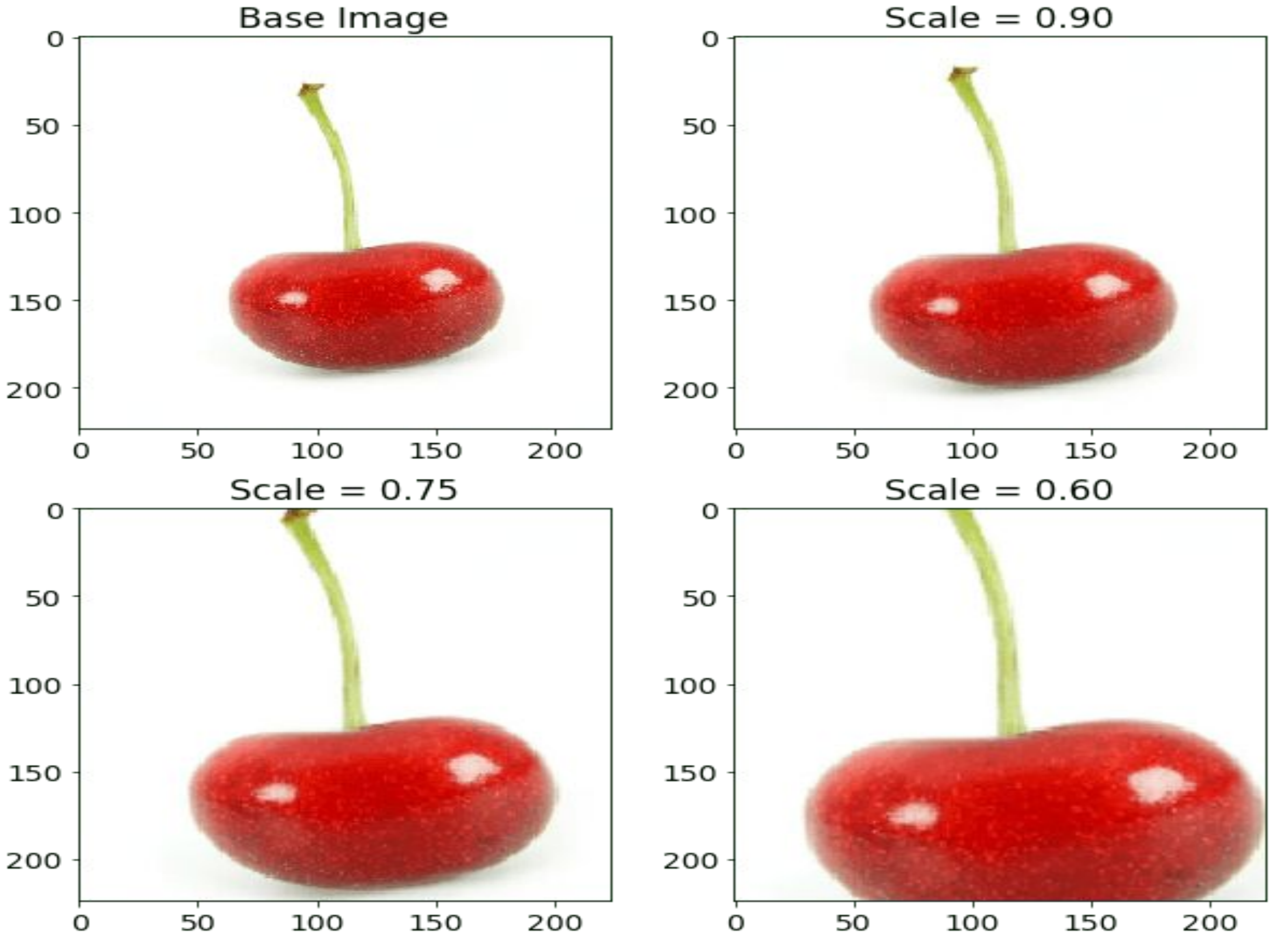
Data Augmentation

- Rotation- The image is rotated by a degree between 0 and 360 degree. Every rotated image will be unique in the model.



Data Augmentation

- Scaling- The image is scaled outward and inward. An object in new image can be smaller or bigger than in the original image by scaling.

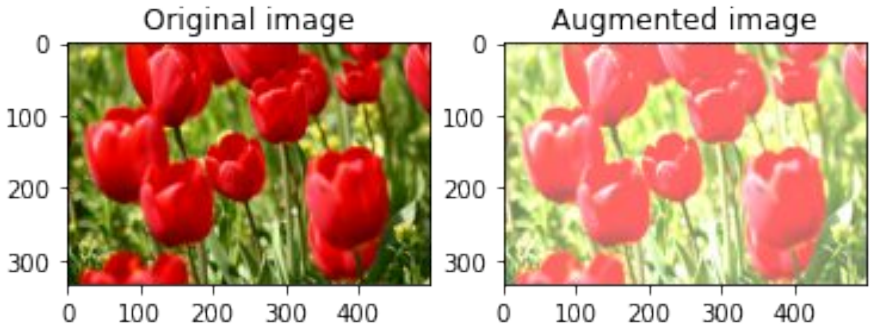


Data Augmentation

- Translation- The image is shifted into various areas along the x-axis or y-axis, so neural network looks everywhere in the image to capture it.

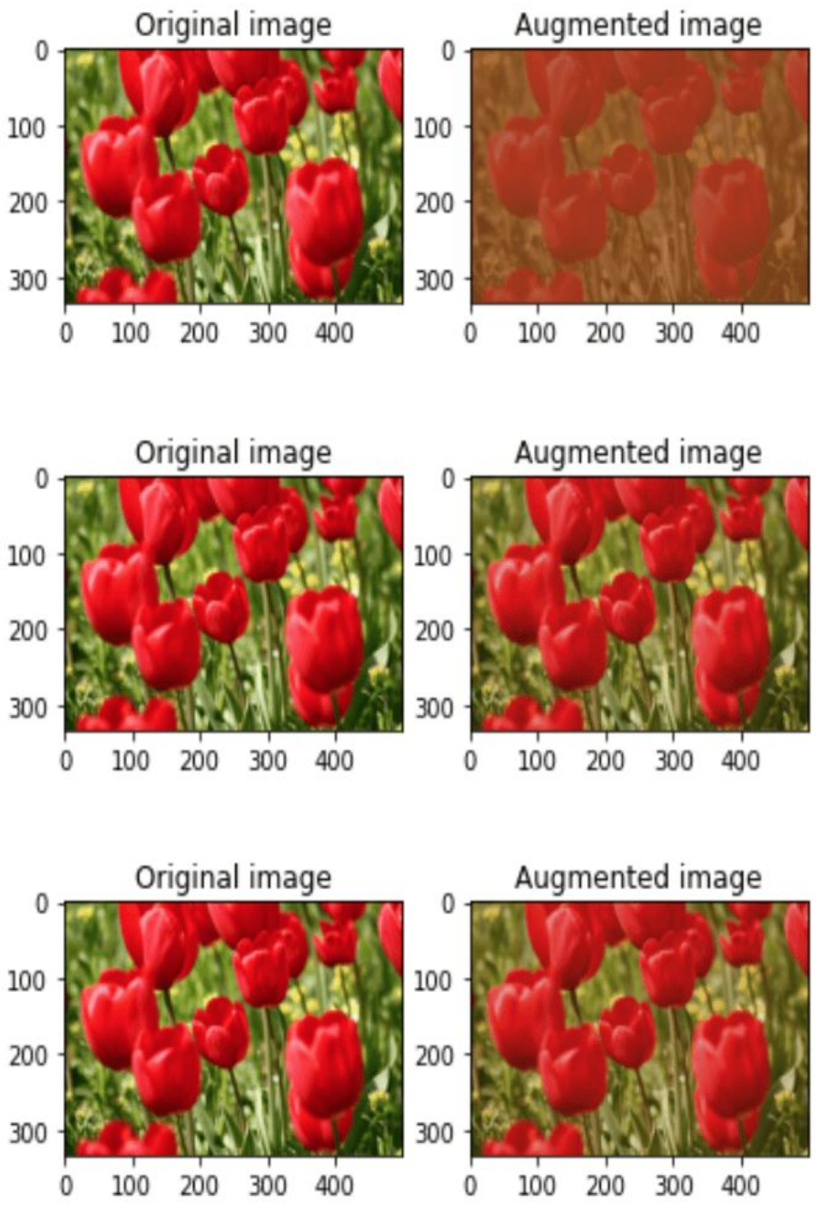
Data Augmentation

- Brightness- The brightness of the image is changed and new image will be darker or lighter. This technique allows the model to recognize image in different lighting levels.



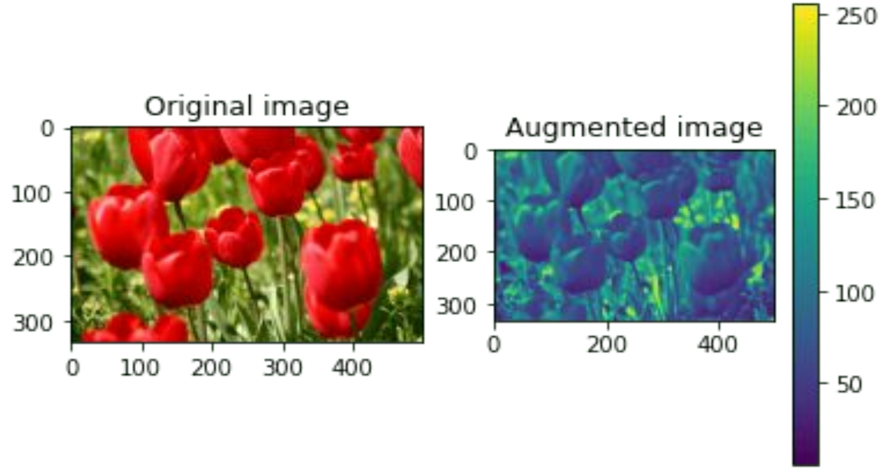
Data Augmentation

- Contrast- The contrast of the image is changed and new image will be different from luminance and colour aspects. The following image's contrast is changed randomly.

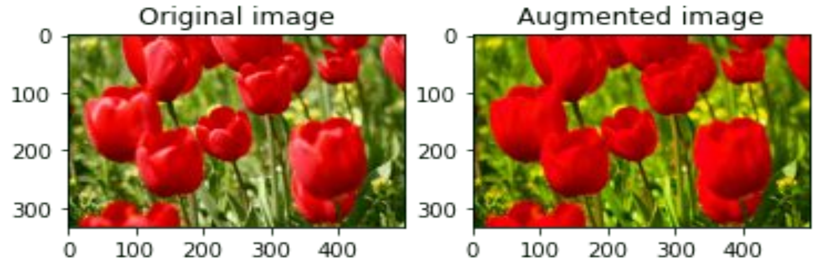


Color Augmentation

- The color of image is changed by new pixel values.



- Saturation is depth or intensity of color in an image.



Data augmentation techniques in NLP models

- Data augmentation techniques are applied on character, word and text levels
- Easy Data Augmentation (EDA) Methods - include easy text transformations,
- Example a word is chosen randomly from the sentence and replaced with one of this word synonyms or two words are chosen and swapped in the sentence.
- EDA techniques examples in NLP processing are

❖ Synonym replacement

❖ Text Substitution (rule-based, ML-based, mask-based and etc.)

❖ Random insertion

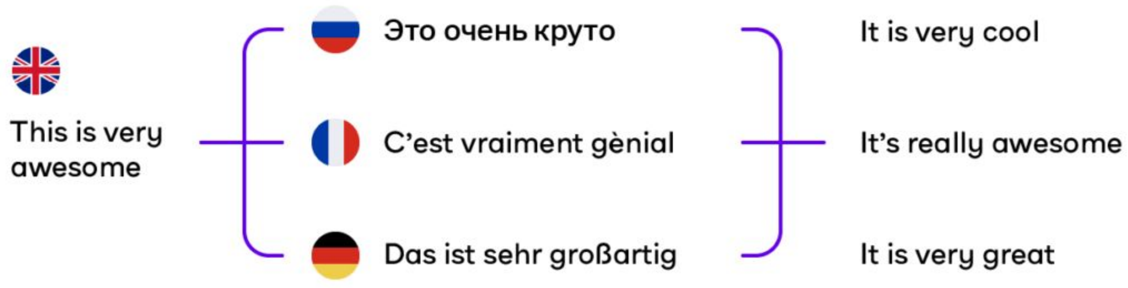
❖ Random swap

❖ Random deletion

❖ Word & sentence shuffling

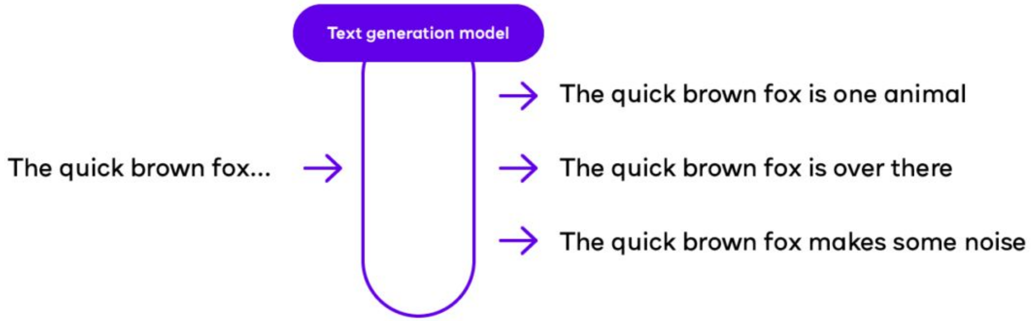
Data Augmentation

- Back Translation- A sentence is translated in one language and then new sentence is translated again in the original language. So, different sentences are created.

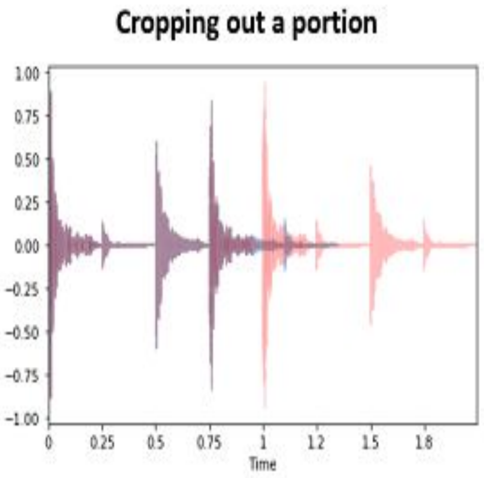


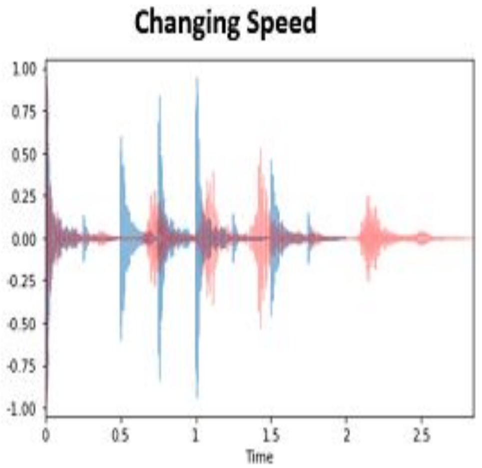
Data Augmentation

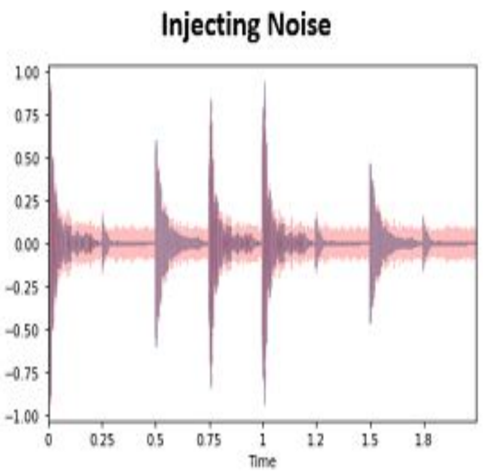
- Text Generation- A generative adversarial networks (GAN) is trained to generate text with a few words.

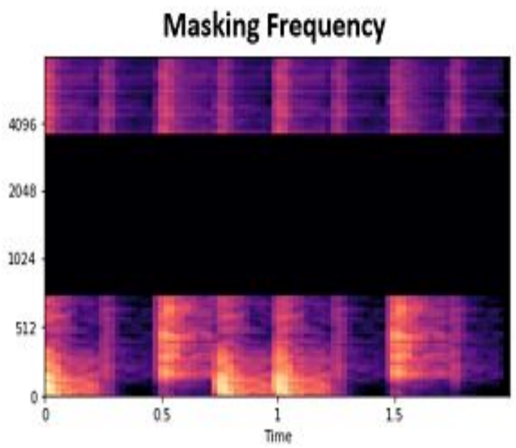


Data augmentation techniques for audio data









- Audio data augmentation methods include

- Cropping out a portion of data, -
- Noise injection,
- Shifting time,
- Speed tuning
- Changing pitch,
- Mixing background noise
- Masking frequency

Regularization

- Regularization is **a technique which makes slight modifications to the learning algorithm such that the model generalizes better**.
- Improves the model's performance on the unseen data as well.

Regularization Techniques in Deep Learning

- L2 & L1 regularization
- These update the general cost function by adding another term known as the regularization term.

Cost function = Loss (say, binary cross entropy) + Regularization term

- It will also reduce overfitting to quite an extent.
- The regularization term differs in L1 and L2.

Regularization Techniques in Deep Learning

- In L2, we have:

$$Cost\ function = Loss + \frac{\lambda}{2m} * \sum \|w\|^2$$

- **lambda** is the regularization parameter.
- It is the hyperparameter whose value is optimized for better results
- L2 regularization is also known as *weight decay* as it forces the weights to decay towards zero (but not exactly zero).

- In L1, we have:

$$Cost\ function = Loss + \frac{\lambda}{2m} * \sum \|w\|$$

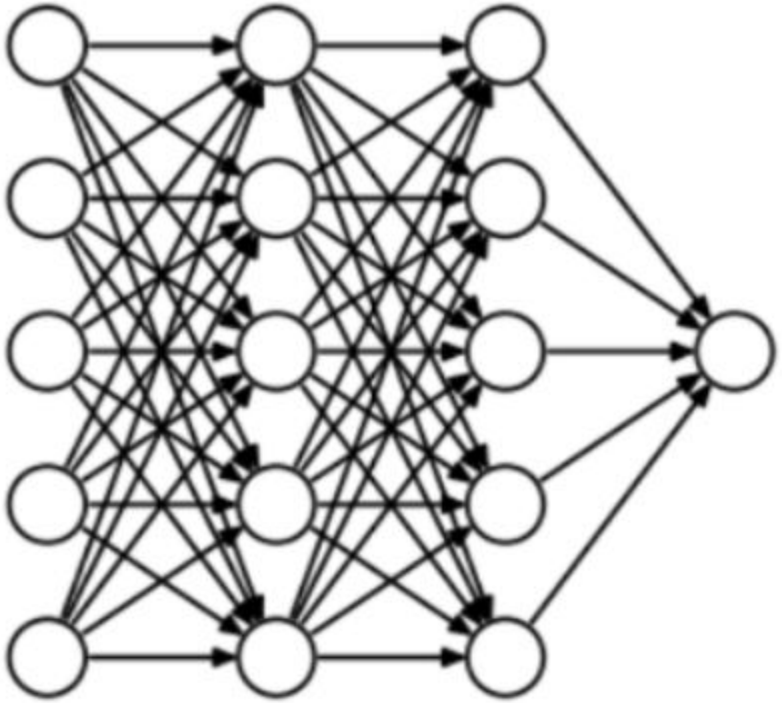
- Penalize the absolute value of the weights.
- Unlike L2, the weights may be reduced to zero here.
- **Hence, it is very useful when we are trying to compress our model. Otherwise, we usually prefer L2 over it.**

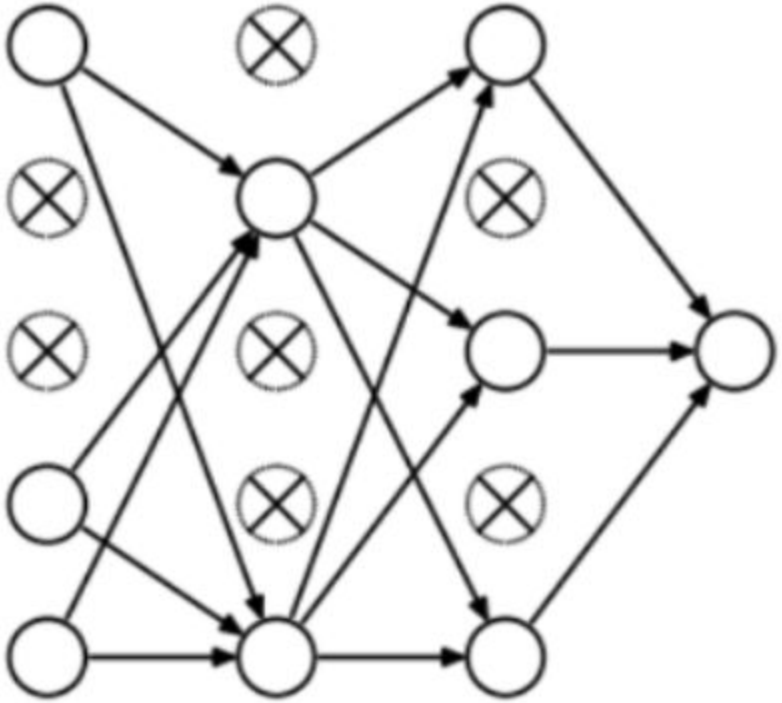
Regularization Techniques

- Dropout

□Most frequently used regularization technique

□At every iteration, it randomly selects some nodes and removes them along with all of their incoming and outgoing connections





Thank You